

Лабораторная работа №4

Работа с программными пакетами

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1. Цели и задачи работы
2. Процесс выполнения лабораторной работы
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1. 1. Цели и задачи работы

1.1 Цель лабораторной работы

Получить практические навыки работы с программными пакетами в Linux.

Изучить использование репозитория, менеджера пакетов **dnf** и утилиты **rpm**.

2. 2. Процесс выполнения лабораторной работы

2.1 Изучаю файлы репозитория

```
mohammedlabsi@mohammedlabsi:~$ su
Password:
root@mohammedlabsi:/home/mohammedlabsi# cd /etc/yum.repos.d/
root@mohammedlabsi:/etc/yum.repos.d# ls
rocky-addons.repo rocky-devel.repo rocky-extras.repo rocky.repo rocky-security.repo
root@mohammedlabsi:/etc/yum.repos.d# cat rocky.repo
# rocky.repo
#
# The mirrorlist system uses the connecting IP address of the client and the
# update status of each mirror to pick current mirrors that are geographically
# close to the client. You should use this for Rocky updates unless you are
# manually picking other mirrors.
#
# If the mirrorlist does not work for you, you can try the commented out
# baseurl line instead.

[baseos]
name=Rocky Linux $releasever - BaseOS
mirrorlist=https://mirrors.rockylinux.org/mirrorlist?arch=$basearch&repo=BaseOS-$releasever$rltype
#baseurl=http://dl.rockylinux.org/$contentdir/$releasever/BaseOS/$basearch/os/
gpgcheck=1
enabled=1
countme=1
metadata_expire=6h
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-Rocky-10

[baseos-debuginfo]
name=Rocky Linux $releasever - BaseOS - Debug
mirrorlist=https://mirrors.rockylinux.org/mirrorlist?arch=$basearch&repo=BaseOS-$releasever-debug$rltype
#baseurl=http://dl.rockylinux.org/$contentdir/$releasever/BaseOS/$basearch/debug/tree/
gpgcheck=1
enabled=0
metadata_expire=6h
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-Rocky-10
```

Рисунок 1: Просмотр каталога репозитория и файла rocky.repo

2.2 Проверяю подключённые репозитории

```
root@mohammedlabsi:/etc/yum.repos.d#  
root@mohammedlabsi:/etc/yum.repos.d# dnf repolist  
repo id                                repo name  
appstream                              Rocky Linux 10 - AppStream  
baseos                                 Rocky Linux 10 - BaseOS  
extras                                 Rocky Linux 10 - Extras  
root@mohammedlabsi:/etc/yum.repos.d# dnf search user  
Last metadata expiration check: 9 days, 19:01:34 ago on Fri 11 Sep 2026 07:09:37 PM MSK.  
===== Name & Summary Matched: user =====  
gnome-user-docs.noarch : GNOME User Documentation  
libuser.x86_64 : A user and group account administration library  
perl-User-pwent.noarch : By-name interface to Perl built-in user name resolver  
samba-usershares.noarch : Provides support for non-root user shares  
usermode.x86_64 : Tools for certain user account management tasks  
userspace-rcu.x86_64 : RCU (read-copy-update) implementation in user-space  
xdg-user-dirs.x86_64 : Handles user special directories  
===== Name Matched: user =====  
gnome-shell-extension-user-theme.noarch : Support for custom themes in GNOME Shell  
xdg-user-dirs-gtk.x86_64 : Gnome integration of special directories  
===== Summary Matched: user =====  
NetworkManager.x86_64 : Network connection manager and user applications  
PackageKit-command-not-found.x86_64 : Ask the user to install command line programs automatically  
accountsservice.x86_64 : D-Bus interfaces for querying and manipulating user account information  
anaconda-gui.x86_64 : Graphical user interface for the Anaconda installer  
anaconda-tui.x86_64 : Textual user interface for the Anaconda installer  
audit.x86_64 : User space tools for kernel auditing  
cachefilesd.x86_64 : CacheFiles user-space management daemon  
cockpit-machines.noarch : Cockpit user interface for virtual machines  
cockpit-ostree.noarch : Cockpit user interface for rpm-ostree  
cockpit-packagekit.noarch : Cockpit user interface for packages  
cockpit-storaged.noarch : Cockpit user interface for storage, using udisks  
console-login-helper-messages.noarch : Combines motd, issue, profile features to show system information to the user before/on  
: login  
criu.x86_64 : Tool for Checkpoint/Restore in User-space  
environment-modules.x86_64 : Provides dynamic modification of a user's environment  
exfatprogs.x86_64 : Userspace utilities for exFAT filesystems  
flatpak-session-helper.x86_64 : User D-Bus service used by flatpak and others  
fuse.x86_64 : File System in Userspace (FUSE) v2 utilities
```

Рисунок 2: Вывод списка репозиторий и поиск пакетов

2.3 Изучаю информацию о пакете nmap

```
root@mohammedlabsi:/etc/yum.repos.d# dnf search nmap
----- Name Exactly Matched: nmap -----
nmap.x86_64 : Network exploration tool and security scanner
----- Name & Summary Matched: nmap -----
nmap-ncat.x86_64 : Nmap's Netcat replacement
root@mohammedlabsi:/etc/yum.repos.d# dnf info nmap
Rocky Linux 10 - BaseOS                                8.4 kB/s | 4.3 kB   00:00
Rocky Linux 10 - BaseOS                                33 MB/s | 29 MB    00:00
Rocky Linux 10 - AppStream                              7.8 kB/s | 4.4 kB   00:00
Rocky Linux 10 - AppStream                             2.5 MB/s | 2.7 MB   00:01
Rocky Linux 10 - Extras                                5.6 kB/s | 3.1 kB   00:00
Available Packages
Name      : nmap
Epoch    : 4
Version   : 7.92
Release   : 5.el10
Architecture : x86_64
Size      : 5.6 M
Source    : nmap-7.92-5.el10.src.rpm
Repository : appstream
Summary   : Network exploration tool and security scanner
URL       : http://nmap.org/
License   : LicenseRef-Nmap
Description : Nmap is a utility for network exploration or security auditing. It supports
: ping scanning (determine which hosts are up), many port scanning techniques
: (determine what services the hosts are offering), and TCP/IP fingerprinting
: (remote host operating system identification). Nmap also offers flexible target
: and port specification, decoy scanning, determination of TCP sequence
: predictability characteristics, reverse-identd scanning, and more. In addition
: to the classic command-line nmap executable, the Nmap suite includes a flexible
: data transfer, redirection, and debugging tool (netcat utility ncat), a utility
: for comparing scan results (ndiff), and a packet generation and response
: analysis tool (nping).

root@mohammedlabsi:/etc/yum.repos.d#
```

Рисунок 3: Поиск пакета nmap и просмотр информации

2.4 Устанавливаю nmap через dnf

```
nmap-7.92-5.el10.x86_64.rpm                                13 MB/s | 5.6 MB   00:00
-----
Total                                                    8.2 MB/s | 5.6 MB   00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing     : nmap-4:7.92-5.el10.x86_64      1/1
  Running scriptlet: nmap-4:7.92-5.el10.x86_64     1/1

Installed:
  nmap-4:7.92-5.el10.x86_64

Complete!
root@mohammedlabsi:/etc/yum.repos.d# dnf install nmap\*
Last metadata expiration check: 0:00:34 ago on Mon 21 Sep 2026 02:12:36 PM MSK.
Package nmap-4:7.92-5.el10.x86_64 is already installed.
Package nmap-ncat-4:7.92-5.el10.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
root@mohammedlabsi:/etc/yum.repos.d# █
```

Рисунок 4: Установка nmap и пакетов по шаблону

2.5 Удаляю nmap

```
Removed:
  nmap-4:7.92-5.el10.x86_64

Complete!
root@mohammedlabsi:/etc/yum.repos.d# dnf remove nmap\*
Dependencies resolved.
=====
Package                        Architecture      Version           Repository        Size
=====
Removing:
  nmap-ncat                  x86_64            4:7.92-5.el10    @AppStream        469 k

Transaction Summary
=====
Remove 1 Package

Freed space: 469 k
Is this ok [y/N]: y
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing                :                               1/1
  Running scriptlet: nmap-ncat-4:7.92-5.el10.x86_64      1/1
  Erasing             : nmap-ncat-4:7.92-5.el10.x86_64    1/1
  Running scriptlet: nmap-ncat-4:7.92-5.el10.x86_64    1/1

Removed:
  nmap-ncat-4:7.92-5.el10.x86_64

Complete!
root@mohammedlabsi:/etc/yum.repos.d# █
```

Рисунок 5: Удаление пакетов nmap

2.6 Просматриваю группы пакетов

```
root@mohammedlabsi:/etc/yum.repos.d# LANG=C dnf groups list
Last metadata expiration check: 0:02:01 ago on Mon Sep 21 14:12:36 2026.
Available Environment Groups:
  Server
  Minimal Install
  Workstation
  Custom Operating System
  Virtualization Host
Installed Environment Groups:
  Server with GUI
Installed Groups:
  Container Management
  Development Tools
  Headless Management
Available Groups:
  Legacy UNIX Compatibility
  Smart Card Support
  Console Internet Tools
  .NET Development
  Graphical Administration Tools
  Network Servers
  RPM Development Tools
  Scientific Support
  Security Tools
  System Tools
root@mohammedlabsi:/etc/yum.repos.d#
```

Рисунок 6: Просмотр доступных групп пакетов

2.7 Устанавливаю RPM Development Tools

```
root@monammedlabsi:/etc/yum.repos.d#
root@monammedlabsi:/etc/yum.repos.d# dnf groups info "RPM Development Tools"
Last metadata expiration check: 0:02:36 ago on Mon 21 Sep 2026 02:12:36 PM MSK.
Group: RPM Development Tools
Description: Tools used for building RPMs, such as rpmbuild.
Mandatory Packages:
  redhat-rpm-config
  rpm-build
Default Packages:
  rpmdevtools
root@monammedlabsi:/etc/yum.repos.d# dnf groupinstall "RPM Development Tools"
Last metadata expiration check: 0:02:48 ago on Mon 21 Sep 2026 02:12:36 PM MSK.
Dependencies resolved.
=====
Package                                Architecture      Version           Repository        Size
=====
Installing group/module packages:
rpmdevtools                            noarch            9.6-9.el10        appstream         87 k
Installing Groups:
RPM Development Tools

Transaction Summary
=====
Install 1 Package

Total download size: 87 k
Installed size: 198 k
Is this ok [y/N]: y
Downloading Packages:
rpmdevtools-9.6-9.el10.noarch.rpm      326 kB/s | 87 kB    00:00
```

Рисунок 7: Информация и установка группы RPM Development Tools

2.8 Изучаю историю dnf

```
root@mohammedlabsi:/etc/yum.repos.d#  
root@mohammedlabsi:/etc/yum.repos.d# dnf history
```

ID	Command line	Date and time	Action(s)	Altered
7	groupinstall RPM Development Tools	2026-09-21 14:15	Install	2
6	remove nmap*	2026-09-21 14:13	Removed	1
5	remove nmap	2026-09-21 14:13	Removed	1
4	install nmap	2026-09-21 14:13	Install	1
3	install gcc make perl bzip2 kernel-devel-6.12.0-211.53.1.el10_2.x86_64	2026-09-11 19:16	D, I	151
2	-y upgrade --nobest	2026-09-11 19:11	I, U	250 E<
1		2026-09-11 19:05	Install	1387 >E

```
root@mohammedlabsi:/etc/yum.repos.d# dnf history undo 6  
Last metadata expiration check: 0:03:27 ago on Mon 21 Sep 2026 02:12:36 PM MSK.  
Dependencies resolved.  
=====
```

Package	Architecture	Version	Repository	Size
Installing:				
nmap-ncat	x86_64	4:7.92-5.el10	appstream	228 k

```
Transaction Summary  
=====
```

Install 1 Package

Рисунок 8: Просмотр истории dnf и отмена транзакции

2.9 Загружаю пакет lynx

```
root@mohammedlabsi:/etc/yum.repos.d#
root@mohammedlabsi:/etc/yum.repos.d# dnf list lynx
Last metadata expiration check: 0:05:37 ago on Mon 21 Sep 2026 02:12:36 PM MSK.
Available Packages
lynx.x86_64                                2.9.0-6.el10                                appstream
root@mohammedlabsi:/etc/yum.repos.d# dnf install lynx --downloadonly
Last metadata expiration check: 0:05:56 ago on Mon 21 Sep 2026 02:12:36 PM MSK.
Dependencies resolved.
=====
Package                Architecture      Version           Repository        Size
=====
Installing:
lynx                    x86_64            2.9.0-6.el10     appstream         1.6 M

Transaction Summary
=====
Install 1 Package

Total download size: 1.6 M
Installed size: 6.0 M
DNF will only download packages for the transaction.
Is this ok [y/N]: y
Downloading Packages:
lynx-2.9.0-6.el10.x86_64.rpm                3.4 MB/s | 1.6 MB    00:00
-----
Total                                         2.1 MB/s | 1.6 MB    00:00
Complete!
The downloaded packages were saved in cache until the next successful transaction.
You can remove cached packages by executing 'dnf clean packages'.
root@mohammedlabsi:/etc/yum.repos.d#
```

Рисунок 9: Загрузка RPM-пакета lynx без установки

2.10 Устанавливаю lynx через rpm

```
root@mohammedlabsi:/etc/yum.repos.d# find /var/cache/dnf/ -name lynx*
/var/cache/dnf/appstream-8f18e4146890fdcf/packages/lynx-2.9.0-6.el10.x86_64.rpm
root@mohammedlabsi:/etc/yum.repos.d# cd /var/cache/dnf/appstream-8f18e4146890fdcf/packages/
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# ls
lynx-2.9.0-6.el10.x86_64.rpm
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# rpm -Uvh lynx-2.9.0-6.el10.x86_64.rpm
Verifying... ##### [100%]
Preparing... ##### [100%]
Updating / installing...
 1:lynx-2.9.0-6.el10 ##### [100%]
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# which lynx
/usr/bin/lynx
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# rpm -qf $(which lynx)
lynx-2.9.0-6.el10.x86_64
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# rpm -qi lynx
Name           : lynx
Version        : 2.9.0
Release        : 6.el10
Architecture   : x86_64
Install Date   : Mon 21 Sep 2026 02:19:31 PM MSK
Group          : Unspecified
Size           : 6283241
License        : GPL-2.0-only
Signature      :
                RSA/SHA256, Thu 22 May 2025 11:34:29 PM MSK, Key ID 5b106c736fedfc85
Source RPM     : lynx-2.9.0-6.el10.src.rpm
Build Date     : Fri 01 Nov 2024 01:03:33 PM MSK
Build Host     : pb-913aa2bb-e2a2-4928-a48c-b613e15933c9-b-x86-64
Packager       : Rocky Linux Build System (Peridot) <releng@rockylinux.org>
Vendor        : Rocky Enterprise Software Foundation
URL            : https://lynx.invisible-island.net/
Summary        : A text-based Web browser
Description    :
Lynx is a text-based Web browser. Lynx does not display any images,
but it does support frames, tables, and most other HTML tags. One
advantage Lynx has over graphical browsers is speed; Lynx starts and
exits quickly and swiftly displays web pages.
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# █
```

Рисунок 10: Установка lynx и просмотр информации о пакете

2.11 Просматриваю файлы пакета lynx

```
root@mohammedlabsi: /var/cache/dnf/appstream-8f18e4146890fd9c/packages# rpm -ql lynx
/etc/lynx-site.cfg
/etc/lynx.cfg
/etc/lynx.lss
/usr/bin/lynx
/usr/lib/.build-id
/usr/lib/.build-id/69
/usr/lib/.build-id/69/febb96205311acd4b77a65d52b8e6f7eea76c3
/usr/share/doc/lynx
/usr/share/doc/lynx/README
/usr/share/doc/lynx/docs
/usr/share/doc/lynx/docs/CHANGES2.3
/usr/share/doc/lynx/docs/CHANGES2.4
/usr/share/doc/lynx/docs/CHANGES2.5
/usr/share/doc/lynx/docs/CHANGES2.6
/usr/share/doc/lynx/docs/CHANGES2.7
/usr/share/doc/lynx/docs/CHANGES2.8
/usr/share/doc/lynx/docs/CRAWL.announce
/usr/share/doc/lynx/docs/FM.announce
/usr/share/doc/lynx/docs/IBMPG-charsets.announce
/usr/share/doc/lynx/docs/README.TRST
/usr/share/doc/lynx/docs/README.chartrans
/usr/share/doc/lynx/docs/README.cookies
/usr/share/doc/lynx/docs/README.defines
/usr/share/doc/lynx/docs/README.metrics
/usr/share/doc/lynx/docs/README.options
/usr/share/doc/lynx/docs/README.unix
```

Рисунок 11: Получение списка файлов пакета lynx

2.12 Просматриваю документацию lynx

```
root@mohammedlabsi: /var/cache/dnf/appstream-8f18e4146890fdcf/packages# rpm -qd lynx
/usr/share/doc/lynx/README
/usr/share/doc/lynx/docs/CHANGES2.3
/usr/share/doc/lynx/docs/CHANGES2.4
/usr/share/doc/lynx/docs/CHANGES2.5
/usr/share/doc/lynx/docs/CHANGES2.6
/usr/share/doc/lynx/docs/CHANGES2.7
/usr/share/doc/lynx/docs/CHANGES2.8
/usr/share/doc/lynx/docs/CRAWL.announce
/usr/share/doc/lynx/docs/FM.announce
/usr/share/doc/lynx/docs/IBMPG-charsets.announce
/usr/share/doc/lynx/docs/README.TRST
/usr/share/doc/lynx/docs/README.chartrans
/usr/share/doc/lynx/docs/README.cookies
/usr/share/doc/lynx/docs/README.defines
/usr/share/doc/lynx/docs/README.metrics
/usr/share/doc/lynx/docs/README.options
/usr/share/doc/lynx/docs/README.release
/usr/share/doc/lynx/docs/README.rootcerts
/usr/share/doc/lynx/docs/README.ssl
/usr/share/doc/lynx/docs/README.sslcerts
/usr/share/doc/lynx/docs/README.sockshtr.announce
/usr/share/doc/lynx/docs/TCPWARE.announce
/usr/share/doc/lynx/docs/VMSWAIS.announce
/usr/share/doc/lynx/docs/djgpp.key
/usr/share/doc/lynx/docs/pdcurse.key
/usr/share/doc/lynx/docs/slang.key
/usr/share/doc/lynx/docs/win-386.announce
/usr/share/doc/lynx/lynx.hlp
/usr/share/doc/lynx/lynx_help/Lynx_users_guide.html
```

Рисунок 12: Получение списка документации пакета lynx

2.13 Изучаю справочную страницу lynx

```
LYNX(1)                                Lynx - the text web browser                                LYNX(1)

NAME
    lynx - a general purpose distributed information browser for the World Wide Web

SYNOPSIS
    lynx [options] [optional paths or URLs]

    lynx [options] [path or URL] -get_data
    data
    --

    lynx [options] [path or URL] -post_data
    data
    --

    Use "lynx -help" to display a complete list of current options.

DESCRIPTION
    Lynx is a fully-featured World Wide Web (WWW) client for users running cursor-addressable, character-cell display devices (e.g., vt100 terminals, vt100 emulators running on Windows or any POSIX platform, or any other "curses-oriented" display). It will display hypertext markup language (HTML) documents containing links to files residing on the local system, as well as files residing on remote systems running Gopher, HTTP, FTP, WAIS, and NNTP servers.

    Lynx has been ported to many systems, including all variants of Unix, Windows (since Windows 95), DOS DJGPP and OS/2, as well as VMS and OS/390.

    Lynx can be used to access information on the World Wide Web, or to build information systems intended primarily for local access. For example, Lynx has been used to build several Campus Wide Information Systems (CWIS). In addition, Lynx can be used to build systems isolated within a single LAN.

OPTIONS
    At start up, Lynx will load any local file or remote URL specified at the command line. For help with URLs, press "?" or "H" while running Lynx. Then follow the link titled, "Help on URLs."

    If more than one local file or remote URL is listed on the command line, Lynx will open only the last interactively. All of the names (local files and remote URLs) are added to the 6)oto history.

Manual page lynx(1) line 1 (press h for help or q to quit)
```

Рисунок 13: Просмотр справочной страницы lynx

2.14 Проверяю запуск lynx

```
Rocky Linux 10                                     Welcome to Rocky Linux

Welcome to Rocky Linux

A community Enterprise Operating System

-----
Rocky Linux is an Enterprise Linux distribution based on the sources available from the CentOS Project and Red Hat, Inc. used for CentOS Stream and Red Hat Enterprise Linux. Rocky Linux intends to be binary compatible with the upstream distributions. The source code can be found at our Rocky Linux Git Service.

Rocky Linux is developed by a diverse team of core developers, with various backgrounds and skill sets that allow them to build, test, and provide Rocky Linux to the world. The core developers are supported by an active user community as well, which includes many testers from the Rocky Linux Testing Team, system administrators, system engineers, managers, core Linux contributors, and Linux enthusiasts.

Rocky Linux has an active user community with many avenues to talk to, work with, or support other Rocky Linux users, as well as responsive communication with the Rocky Linux core development team. There are many ways to join and take part in the community: Mattermost, IRC/Matrix, Forums, Email Lists, Reddit

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Commands: Use arrow keys to move, '?' for help, 'q' to quit, '<-' to go back.
Arrow keys: Up and Down to move. Right to follow a link; Left to go back.
H)elp O)ptions P)rint O)o M)ain screen Q)uit /-search [delete]-history list
```

Рисунок 14: Запуск текстового браузера lynx

2.15 Удаляю lynx

```
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# man lynx
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fdcf/packages#
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# rpm -qc lynx
/etc/lynx-site.cfg
/etc/lynx.cfg
/etc/lynx.lss
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# rpm -q --scripts lynx
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# rpm -e lynx
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# ls
lynx-2.9.0-6.el10.x86_64.rpm
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fdcf/packages#
```

Рисунок 15: Просмотр конфигурации, скриптов и удаление lynx

2.16 Проверяю пакет dnsmasq

```
root@mohammedlabsi: /var/cache/dnf/appstream-8f18e4146890fd9c/packages# dnf list dnsmasq
Last metadata expiration check: 0:12:49 ago on Mon 21 Sep 2026 02:12:36 PM MSK.
Installed Packages
dnsmasq.x86_64                                2.90-7.el10_2                                @AppStream
root@mohammedlabsi: /var/cache/dnf/appstream-8f18e4146890fd9c/packages# dnf install dnsmasq-utils-2.90-7.el10_2.x86_64
Last metadata expiration check: 0:12:59 ago on Mon 21 Sep 2026 02:12:36 PM MSK.
Dependencies resolved.
=====
Package                                Architecture      Version           Repository        Size
=====
Installing:
dnsmasq-utils                          x86_64            2.90-7.el10_2     appstream         39 k
=====

Transaction Summary
=====
Install 1 Package

Total download size: 39 k
Installed size: 101 k
Is this ok [y/N]: y
Downloading Packages:
dnsmasq-utils-2.90-7.el10_2.x86_64.rpm                                103 kB/s | 39 kB    00:00
-----
Total                                                                    62 kB/s | 39 kB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      : dnsmasq-utils-2.90-7.el10_2.x86_64                1/1
  Installing     : dnsmasq-utils-2.90-7.el10_2.x86_64                1/1
  Running scriptlet: dnsmasq-utils-2.90-7.el10_2.x86_64                1/1

Installed:
dnsmasq-utils-2.90-7.el10_2.x86_64
```

Рисунок 16: Проверка dnsmasq и установка дополнительных утилит

2.17 Получаю информацию о dnsmasq

```
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# which dnsmasq
/usr/sbin/dnsmasq
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# rpm -qf $(which dnsmasq)
dnsmasq-2.90-7.el10_2.x86_64
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# rpm -qi dnsmasq
Name       : dnsmasq
Version    : 2.90
Release    : 7.el10_2
Architecture: x86_64
Install Date: Fri 11 Sep 2026 07:06:39 PM MSK
Group      : Unspecified
Size       : 794823
License    : GPL-2.0-only OR GPL-3.0-only
Signature  :
            RSA/SHA256, Thu 21 May 2026 07:23:08 AM MSK, Key ID 5b106c736fedfc85
Source RPM : dnsmasq-2.90-7.el10_2.src.rpm
Build Date : Wed 20 May 2026 01:13:21 PM MSK
Build Host : ord1-prod-x86build008.svc.aws.rockylinux.org
Packager   : Rocky Linux Build System <releng@rockylinux.org>
Vendor     : Rocky Enterprise Software Foundation
VCS        : git:git://thekelleys.org.uk/dnsmasq.git
URL        : http://www.thekelleys.org.uk/dnsmasq/
Bug URL    : https://bugs.rockylinux.org
Summary    : A lightweight DHCP/caching DNS server
Description:
Dnsmasq is lightweight, easy to configure DNS forwarder and DHCP server.
It is designed to provide DNS and, optionally, DHCP, to a small network.
It can serve the names of local machines which are not in the global
DNS. The DHCP server integrates with the DNS server and allows machines
with DHCP-allocated addresses to appear in the DNS with names configured
either in each host or in a central configuration file. Dnsmasq supports
static and dynamic DHCP leases and BOOTP for network booting of disk-less
machines.
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# █
```

Рисунок 17: Определение принадлежности файла пакеты dnsmasq

2.18 Просматриваю файлы dnsmasq

```
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fd9c/packages# rpm -ql dnsmasq
/etc/dbus-1/system.d/dnsmasq.conf
/etc/dnsmasq.conf
/etc/dnsmasq.d
/usr/lib/.build-id
/usr/lib/.build-id/b9
/usr/lib/.build-id/b9/4ac4ddee2d193794c27a8ace2f76993101ddce
/usr/lib/systemd/system/dnsmasq.service
/usr/lib/sysusers.d/dnsmasq.conf
/usr/lib/tmpfiles.d/dnsmasq.conf
/usr/sbin/dnsmasq
/usr/share/dnsmasq
/usr/share/dnsmasq/trust-anchors.conf
/usr/share/doc/dnsmasq
/usr/share/doc/dnsmasq/CHANGELOG
/usr/share/doc/dnsmasq/DBus-interface
/usr/share/doc/dnsmasq/FAQ
/usr/share/doc/dnsmasq/doc.html
/usr/share/doc/dnsmasq/setup.html
/usr/share/licenses/dnsmasq
/usr/share/licenses/dnsmasq/COPYING
/usr/share/licenses/dnsmasq/COPYING-v3
/usr/share/man/man8/dnsmasq.8.gz
/var/lib/dnsmasq
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fd9c/packages# rpm -qd dnsmasq
/usr/share/doc/dnsmasq/CHANGELOG
/usr/share/doc/dnsmasq/DBus-interface
/usr/share/doc/dnsmasq/FAQ
/usr/share/doc/dnsmasq/doc.html
/usr/share/doc/dnsmasq/setup.html
/usr/share/man/man8/dnsmasq.8.gz
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fd9c/packages# █
```

Рисунок 18: Просмотр файлов и документации пакета dnsmasq

2.19 Изучаю справочную страницу dnsmasq

DNSMASQ(8)	System Manager's Manual	DNSMASQ(8)
NAME		
dnsmasq - A lightweight DHCP and caching DNS server.		
SYNOPSIS		
dnsmasq <u>[OPTION]...</u>		
DESCRIPTION		
dnsmasq is a lightweight DNS, TFTP, PXE, router advertisement and DHCP server. It is intended to provide coupled DNS and DHCP service to a LAN. Dnsmasq accepts DNS queries and either answers them from a small, local, cache or forwards them to a real, recursive, DNS server. It loads the contents of /etc/hosts so that local hostnames which do not appear in the global DNS can be resolved and also answers DNS queries for DHCP configured hosts. It can also act as the authoritative DNS server for one or more domains, allowing local names to appear in the global DNS. It can be configured to do DNSSEC validation. The dnsmasq DHCP server supports static address assignments and multiple networks. It automatically sends a sensible default set of DHCP options, and can be configured to send any desired set of DHCP options, including vendor-encapsulated options. It includes a secure, read-only, TFTP server to allow net/PXE boot of DHCP hosts and also supports BOOTP. The PXE support is full featured, and includes a proxy mode which supplies PXE information to clients whilst DHCP address allocation is done by another server. The dnsmasq DHCPv6 server provides the same set of features as the DHCPv4 server, and in addition, it includes router advertisements and a neat feature which allows naming for clients which use DHCPv4 and stateless autoconfiguration only for IPv6 configuration. There is support for doing address allocation (both DHCPv6 and RA) from subnets which are dynamically delegated via DHCPv6 prefix delegation. Dnsmasq is coded with small embedded systems in mind. It aims for the smallest possible memory footprint compatible with the supported functions, and allows unneeded functions to be omitted from the compiled binary.		
OPTIONS		
Note that in general missing parameters are allowed and switch off functions, for instance "--pid-file" disables writing a PID file. On BSD, unless the GNU getopt library is linked, the long form of the options does not work on the command line; it is still recognised in the configuration file. --test Read and syntax check configuration file(s). Exit with code 0 if all is OK, or a non-zero code otherwise. Do		
Manual page dnsmasq(8) line 1 (press h for help or q to quit)		

Рисунок 19: Просмотр руководства dnsmasq

2.20 Анализирую скрипты dnsmasq

```
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fd9c/packages# rpm -qc dnsmasq
/etc/dbus-1/system.d/dnsmasq.conf
/etc/dnsmasq.conf
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fd9c/packages# rpm -q --scripts dnsmasq
preinstall scriptlet (using /bin/sh):
#precreate users so that rpm can install files owned by that user

# generated from dnsmasq-systemd-sysusers.conf
getent group 'dnsmasq' >/dev/null || groupadd -r 'dnsmasq' || :
getent passwd 'dnsmasq' >/dev/null || \
    useradd -r -g 'dnsmasq' -d '/var/lib/dnsmasq' -s '/usr/sbin/nologin' -c 'Dnsmasq DHCP and DNS server' 'dnsmasq' || :
postinstall scriptlet (using /bin/sh):

if [ $1 -eq 1 ] && [ -x "/usr/lib/systemd/systemd-update-helper" ]; then
    # Initial installation
    /usr/lib/systemd/systemd-update-helper install-system-units dnsmasq.service || :
fi
preuninstall scriptlet (using /bin/sh):

if [ $1 -eq 0 ] && [ -x "/usr/lib/systemd/systemd-update-helper" ]; then
    # Package removal, not upgrade
    /usr/lib/systemd/systemd-update-helper remove-system-units dnsmasq.service || :
fi
postuninstall scriptlet (using /bin/sh):

if [ $1 -ge 1 ] && [ -x "/usr/lib/systemd/systemd-update-helper" ]; then
    # Package upgrade, not uninstall
    /usr/lib/systemd/systemd-update-helper mark-restart-system-units dnsmasq.service || :
fi
root@mohammedlabsi:/var/cache/dnf/appstream-8f18e4146890fd9c/packages#
```

Рисунок 20: Просмотр конфигурационных файлов и RPM-скриптов dnsmasq

3. 3. Выводы по проделанной работе

В ходе лабораторной работы были изучены репозитории Rocky Linux и файлы их настройки в каталоге **/etc/yum.repos.d**.

Были освоены основные возможности **dnf**: просмотр репозиториев, поиск, установка, удаление пакетов, работа с группами пакетов и просмотр истории транзакций.

Также была изучена работа с **rpm**: установка пакета из локального RPM-файла, просмотр информации о пакете, определение принадлежности файла пакету, просмотр документации, конфигурационных файлов и установочных скриптов.

На практике были рассмотрены пакеты **nmap**, **lynx** и **dnsmasq**.